

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

AIM: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE: -

```
print("DIFFIE-HELLMAN KEY EXCHANGE")
print("-----")

p = int(input("Enter prime number (p): "))
g = int(input("Enter primitive root (g): "))

a = int(input("Enter Tanu's private key: "))
b = int(input("Enter Tanishka's private key: "))

A = pow(g, a, p)

B = pow(g, b, p)

key_Tanu = pow(B, a, p)
key_Tanishka = pow(A, b, p)

print("\n--- RESULTS ---")

print("Tanu's Public Key:", A)
print("Tanishka's Public Key:", B)

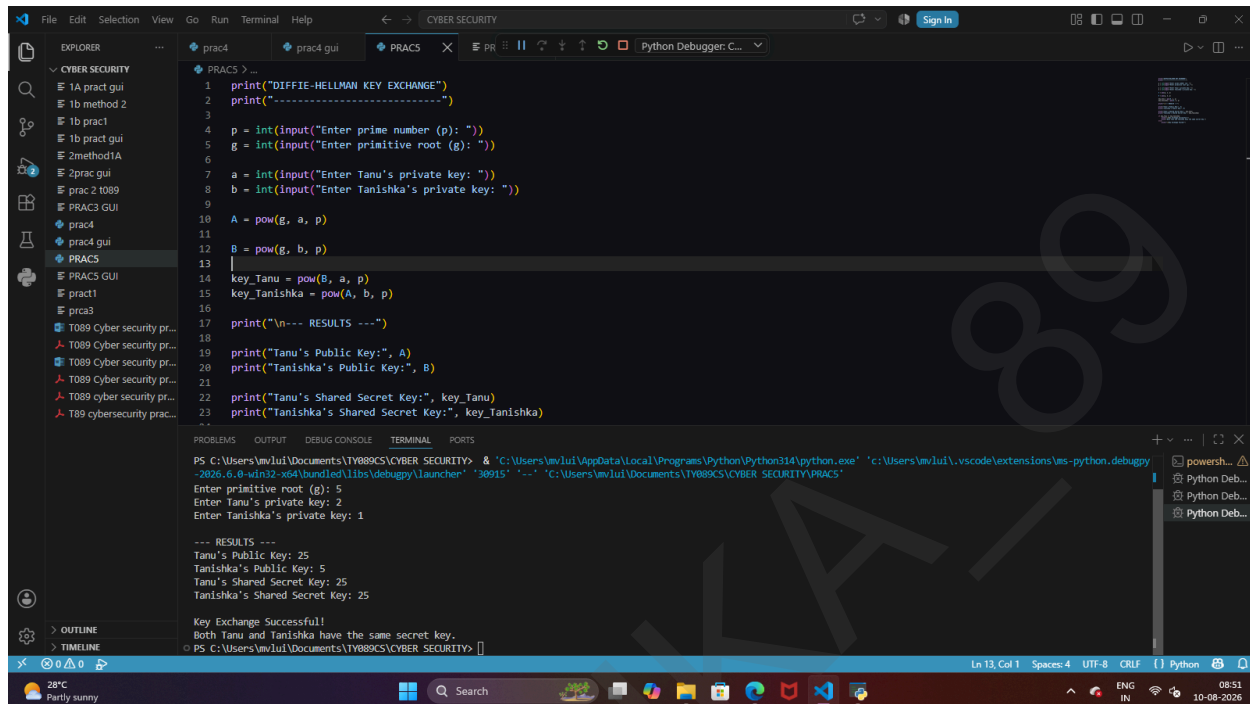
print("Tanu's Shared Secret Key:", key_Tanu)
print("Tanishka's Shared Secret Key:", key_Tanishka)

if key_Tanu == key_Tanishka:
    print("\nKey Exchange Successful!")
    print("Both Tanu and Tanishka have the same secret key.")
else:
    print("\nKey Exchange Failed!")
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

OUTPUT:-



```
1 print("DIFFIE-HELLMAN KEY EXCHANGE")
2 print("-----")
3
4 p = int(input("Enter prime number (p): "))
5 g = int(input("Enter primitive root (g): "))
6
7 a = int(input("Enter Tanu's private key: "))
8 b = int(input("Enter Tanishka's private key: "))
9
10 A = pow(g, a, p)
11
12 B = pow(g, b, p)
13
14 key_Tanu = pow(B, a, p)
15 key_Tanishka = pow(A, b, p)
16
17 print("\n--- RESULTS ---")
18
19 print("Tanu's Public Key:", A)
20 print("Tanishka's Public Key:", B)
21
22 print("Tanu's Shared Secret Key:", key_Tanu)
23 print("Tanishka's Shared Secret Key:", key_Tanishka)
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\vm\lul\Documents\TY089CS\CYBER SECURITY> & "C:\Users\vm\lul\AppData\Local\Programs\Python\Python314\python.exe" "c:\Users\vm\lul\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher" "30915" "-..." "C:\Users\vm\lul\Documents\TY089CS\CYBER SECURITY\PRACS"
Enter primitive root (g): 5
Enter Tanu's private key: 2
Enter Tanishka's private key: 1

--- RESULTS ---
Tanu's Public Key: 25
Tanishka's Public Key: 5
Tanu's Shared Secret Key: 25
Tanishka's Shared Secret Key: 25

Key Exchange Successful!
Both Tanu and Tanishka have the same secret key.
PS C:\Users\vm\lul\Documents\TY089CS\CYBER SECURITY>
```

GUI:-

```
import tkinter as tk
from tkinter import messagebox

def calculate_keys():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(Riddhi_entry.get())
        b = int(Tanu_entry.get())

        A = pow(g, a, p)
        B = pow(g, b, p)

        key_Riddhi = pow(B, a, p)
        key_Tanu = pow(A, b, p)

        Riddhi_public_label.config(
            text="Riddhi's Public Key: " + str(A)
        )
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

```
Tanu_public_label.config(
    text="Tanu's Public Key: " + str(B)
)

Riddhi_shared_label.config(
    text="Riddhi's Shared Secret Key: " + str(key_Riddhi)
)

Tanu_shared_label.config(
    text="Tanu's Shared Secret Key: " + str(key_Tanu)
)

if key_Riddhi == key_Tanu:
    result_label.config(
        text="Key Exchange Successful!\n"
        "Both Riddhi and Tanu have the same secret key.",
        fg="green"
    )
else:
    result_label.config(
        text="Key Exchange Failed!",
        fg="red"
    )

except ValueError:
    messagebox.showerror(
        "Error",
        "Please enter valid numbers."
    )

def clear_all():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    Riddhi_entry.delete(0, tk.END)
    Tanu_entry.delete(0, tk.END)

    Riddhi_public_label.config(text="")
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

```
Tanu_public_label.config(text="")
Riddhi_shared_label.config(text="")
Tanu_shared_label.config(text="")
result_label.config(text="")

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("650x650")
root.resizable(False, False)

title = tk.Label(
    root,
    text="DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 20, "bold")
)
title.pack(pady=20)

tk.Label(
    root,
    text="Enter Prime Number (p):",
    font=("Arial", 12, "bold")
).pack()

p_entry = tk.Entry(
    root,
    width=30,
    font=("Arial", 12)
)
p_entry.pack(pady=5)

tk.Label(
    root,
    text="Enter Primitive Root (g):",
    font=("Arial", 12, "bold")
).pack()

g_entry = tk.Entry(
    root,
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

```
        width=30,  
        font=("Arial", 12)  
    )  
    g_entry.pack(pady=5)  
  
    tk.Label(  
        root,  
        text="Enter Riddhi's Private Key:",  
        font=("Arial", 12, "bold")  
    ).pack()  
  
    Riddhi_entry = tk.Entry(  
        root,  
        width=30,  
        font=("Arial", 12)  
    )  
    Riddhi_entry.pack(pady=5)  
  
    tk.Label(  
        root,  
        text="Enter Tanu's Private Key:",  
        font=("Arial", 12, "bold")  
    ).pack()  
  
    Tanu_entry = tk.Entry(  
        root,  
        width=30,  
        font=("Arial", 12)  
    )  
    Tanu_entry.pack(pady=5)  
  
    tk.Button(  
        root,  
        text="Generate Shared Key",  
        command=calculate_keys,  
        font=("Arial", 11, "bold"),  
        padx=20,  
        pady=7
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

```
) .pack(pady=15)

Riddhi_public_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)
Riddhi_public_label.pack(pady=5)

Tanu_public_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)
Tanu_public_label.pack(pady=5)

Riddhi_shared_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)
Riddhi_shared_label.pack(pady=5)

Tanu_shared_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold")
)
Tanu_shared_label.pack(pady=5)

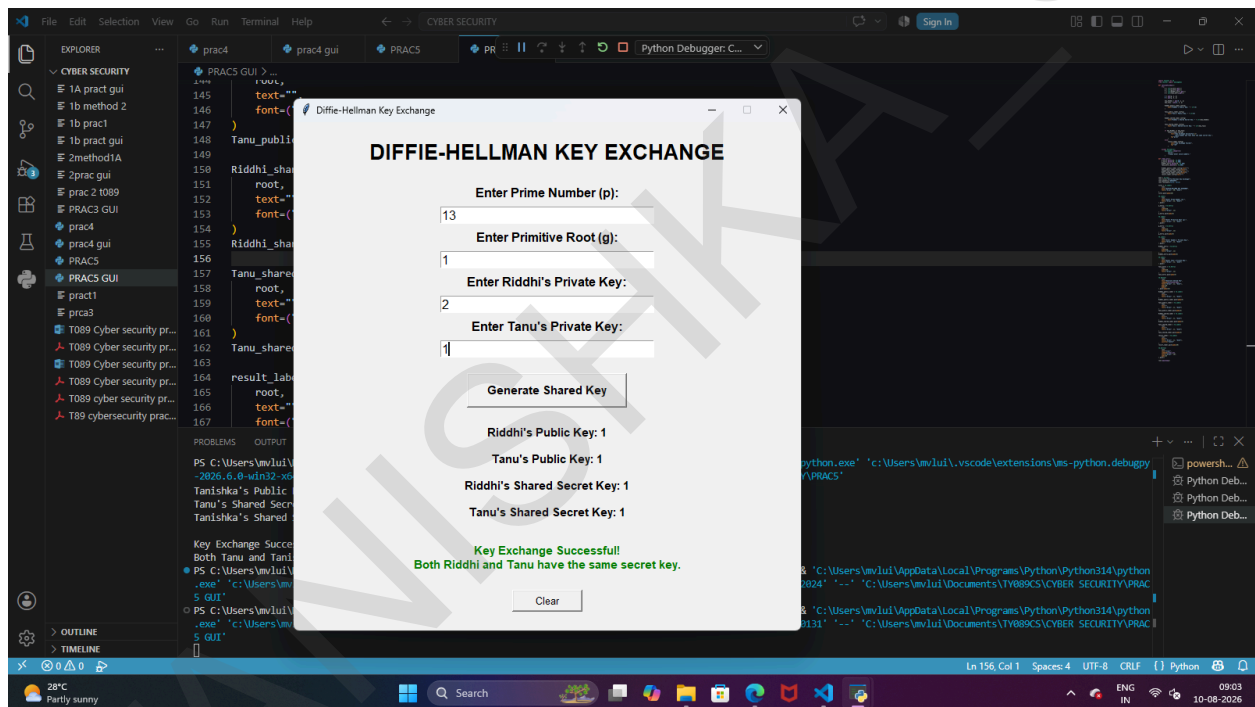
result_label = tk.Label(
    root,
    text="",
    font=("Arial", 11, "bold"),
    justify="center"
)
result_label.pack(pady=20)
```

NAME: TANISHKA KUTWAL
ROLL NO: T089

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT: CYBER SECURITY
PRACTICAL NO: 5

```
tk.Button(  
    root,  
    text="Clear",  
    command=clear_all,  
    font=("Arial", 10),  
    padx=25  
) .pack()  
  
root.mainloop()
```

OUTPUT:



NAME: TANISHKA KUTWAL
ROLL NO: T089